



Machine Learning

MCA 193

UNIT 1: INTRODUCTION TO MACHINE LEARNING

1.1 Introduction: Machine learning (ML) is a subdomain of artificial intelligence (AI) that focuses on developing systems that learn or improve performance based on the data they ingest. Machine learning is a growing technology which enables computers to learn automatically from past data. Machine learning uses various algorithms for building mathematical models and making predictions using historical data or information. Currently, it is being used for various tasks such as image recognition, speech recognition, email filtering, Facebook auto-tagging, recommender system and many more.

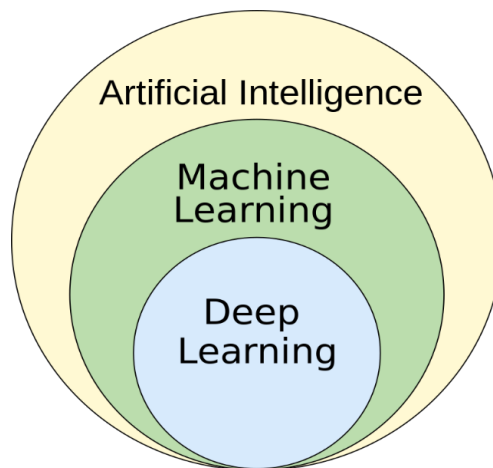
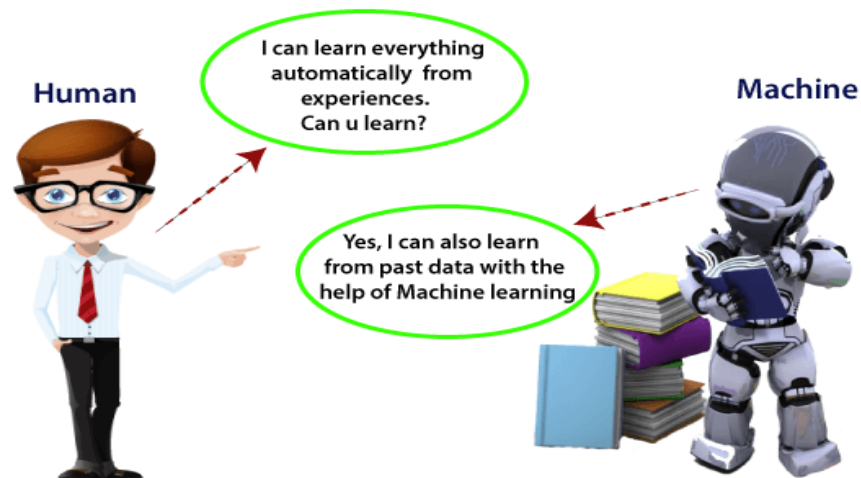


Fig: Artificial Intelligence, Machine Learning and Deep Learning

Artificial intelligence is a broad word that refers to systems or machines that resemble human intelligence. Machine learning and AI are frequently discussed together, and the terms are occasionally used interchangeably, although they do not signify the same thing. A crucial distinction is that, while all machine learning is AI, not all AI is machine learning.



{Machine learning (ML) is a part of artificial intelligence and growing technology that enables machines to learn from past data and perform a given task automatically. Machine Learning allows the computers to learn from the experiences by its own, use statistical methods to improve the performance and predict the output without being explicitly programmed. The popular applications of ML are Email spam filtering, product recommendations, online fraud detection, etc.}

How does Machine Learning work?

The term machine learning was first introduced by Arthur Samuel in 1959. We can define it in a summarized way as: Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed. A machine learning system builds

prediction models, learns from previous data, and predicts the output of new data whenever it receives it. The amount of data helps to build a better model that accurately predicts the output, which in turn affects the accuracy of the predicted output.

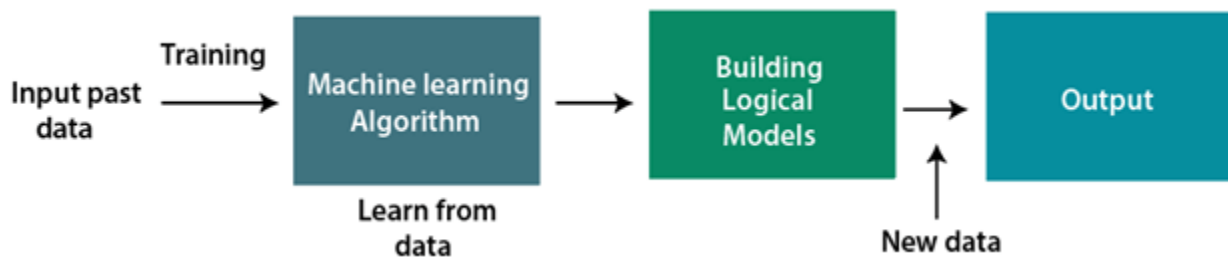


Fig: Block diagram of ML Algorithm's

Features of Machine learning

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.

Importance of Machine Learning

- Rapid increment in the production of data.
- Solving complex problems, which are difficult for a human.
- Decision making in various sector including finance.
- Finding hidden patterns and extracting useful information from data.

1.2 Components of machine learning

Machine learning encompasses several key components that work together to enable systems to learn from data and make predictions or decisions without explicit programming. These components include:

- 1. Data:** High-quality, relevant, and sufficient data is essential for training machine learning models. This includes both input data (features) and output data (labels or target variables), which the model uses to learn patterns and make predictions.
- 2. Feature representation:** Features are the individual measurable properties or characteristics of the data that are used as input for the machine learning model. Feature representation involves selecting, extracting, and transforming the raw data into a format that the model can understand and learn from effectively.
- 3. Model architecture:** The model architecture defines the structure and behavior of the machine learning model. It includes the type of algorithm or learning approach used (e.g., decision trees, neural networks, support vector machines) and the specific parameters and configurations that govern how the model learns from the data.
- 4. Training algorithm:** The training algorithm is responsible for adjusting the parameters of the model to minimize the difference between the model's predictions and the actual output (labels) in the training data. This typically involves an optimization process that iteratively updates the model's parameters based on the training data.
- 5. Loss function:** The loss function measures the difference between the model's predictions and the actual output (labels) in the training data. It provides a quantitative measure of the model's performance during training and is used by the optimization algorithm to adjust the model's parameters.
- 6. Evaluation metrics:** Evaluation metrics are used to assess the performance of the trained machine learning model on unseen data. Common evaluation metrics include accuracy, precision, recall, F1-score, and area under the ROC curve (AUC), depending on the specific task and requirements.
- 7. Validation strategy:** Validation strategies, such as cross-validation or holdout validation, are used to assess the generalization performance of the model and to

tune hyper parameters. These strategies involve partitioning the data into training and validation subsets to simulate how the model will perform on unseen data.

8. Hyper parameters: Hyper parameters are parameters that are not learned from the data but instead control the behavior and performance of the machine learning model. Examples include the learning rate, regularization strength, and model complexity parameters.

9. Deployment and monitoring: Once trained, machine learning models need to be deployed into production environments where they can make predictions or decisions on new data. Deployment involves integrating the model into applications or systems, while monitoring involves continuously evaluating the model's performance and updating it as necessary. These components collectively form the foundation of machine learning systems and are essential for building effective and reliable models for various applications.

1.3 Learning models

A **learning model** in machine learning refers to an algorithm or function that learns from data and makes predictions or decisions. It identifies patterns and relationships within datasets without being explicitly programmed. Learning models are broadly classified into **supervised learning, unsupervised learning and reinforcement learning**.

1. Supervised learning: In supervised learning, models are trained on labeled datasets. Each data point has input features (independent variables) and a corresponding output label (dependent variable). Supervised learning model is further divided into:

Classification: Used for predicting categorical labels or classes. (e.g., spam detection), and

Regression: Used for predicting continuous values. (e.g., house price prediction).

Algorithms: Linear Regression, Logistic Regression, Decision Trees, Support Vector Machines (SVM), Neural Networks, Random Forests and Naive Bayes.

2. **Unsupervised learning:** Unsupervised models process raw, unlabeled datasets. They discover patterns and structures without explicit output labels.

The model learns from unlabeled data, identifying patterns and structures.

- **Clustering:** Grouping similar data points together based on some similarity metric. Examples include customer segmentation, Dimensionality Reduction (e.g., PCA). Algorithms: K-Means, DBSCAN, Hierarchical clustering.
- **Association:** Association rule mining finds interesting associations and relationships among large sets of data items. This rule shows how frequently a item set occurs in a transaction. A typical example is a Market Based Analysis. Market Based Analysis is one of the key techniques used by large relations to show associations between items. Algorithms include Apriori, FB-Growth, ECLAT, etc.

3. **Reinforcement learning:** This model learns by interacting with an environment and receiving rewards or penalties. Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences and improving its performance. Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards. In reinforcement learning, there is no labelled data like supervised learning, and agents learn from their experiences only. Examples includes Game playing (AlphaGo), Robotics, Self-driving cars. Algorithm: Q-learning, deep Q-networks (DQN) and policy gradient methods.

Types of learning models

i) Probabilistic models

Probabilistic models use probability theory to make predictions and handle uncertainty in data. These models estimate the likelihood of different outcomes and are widely used in fields like natural language processing, speech recognition and decision-making under uncertainty. Probabilistic models are an essential component of machine learning, which aims to learn patterns from data and make predictions on new, unseen data. Probabilistic models are used in various applications such as image and speech recognition, natural language processing and recommendation systems. In recent years, significant progress has been made in developing probabilistic models that can handle large datasets efficiently.

Categories of probabilistic models

These models can be classified into the following categories:

- ✚ **Generative models:** Generative models aim to model the joint distribution of the input and output variables. These models generate new data based on the probability distribution of the original dataset. Generative models are powerful because they can generate new data that resembles the training data. They can be used for tasks such as image and speech synthesis, language translation and text generation.
- ✚ **Discriminative models:** The discriminative model aims to model the conditional distribution of the output variable given the input variable. They learn a decision boundary that separates the different classes of the output variable. Discriminative models are useful when the focus is on making accurate predictions rather than generating new data. They can be used for tasks such as image recognition, speech recognition and sentiment analysis.

- ✚ **Graphical Models:** These models use graphical representations to show the conditional dependence between variables. They are commonly used for tasks such as image recognition, natural language processing, and causal inference.

Importance of Probabilistic Models

- Probabilistic models play a crucial role in the field of machine learning, providing a framework for understanding the underlying patterns and complexities in massive datasets.
- Probabilistic models provide a natural way to reason about the likelihood of different outcomes and can help us understand the underlying structure of the data.
- Probabilistic models help enable researchers and practitioners to make informed decisions when faced with uncertainty.
- Probabilistic models allow us to perform Bayesian inference, which is a powerful method for updating our beliefs about a hypothesis based on new data. This can be particularly useful in situations where we need to make decisions under uncertainty.

Applications of probabilistic models

- Spam filtering (Naïve Bayes)
- Speech recognition (HMMs)
- Anomaly detection (GMMs)
- ◀ - Weather prediction (Bayesian networks)
- Recommender systems (Probabilistic matrix factorization)

ii) Geometric models: A geometric model can be a mathematical model of a system or element that uses geometry for the explanation of its properties and connections. In geometric models, features could be described as points in two dimensions (x- and y-axis) or a three-dimensional space (x, y, and z). Even when features are not

intrinsically geometric, they could be modelled in a geometric manner (for example, temperature as a function of time can be modelled in two axes).

In geometric models, there are two ways we could impose similarity.

- ✓ We could use geometric concepts like lines or planes to segment (classify) the instance space. These are called **linear models**.
- ✓ Alternatively, we can use the geometric notion of distance to represent similarity. In this case, if two points are close together, they have similar values for features and thus can be classed as similar. We call such models as **distance based models**.
- ✓ The Following are a few examples of geometric models that are used commonly in machine learning:
 - **Convolutional Neural Networks (CNNs)**: CNNs are often used for recognizing images and machine learning tasks. They use convolutional layers, which apply filters to input data to capture local relationships and spatial patterns.
 - **Graph Neural Networks (GNNs)**: GNNs have been developed to analyze data presented as charts, which are composed of nodes and edges that indicate objects and the connections between each other.
 - **Support Vector Machines (SVMs)** are supervised learning models that are employed for classification and regression problems. Finding the best hyperplane that distinguishes between several classes or fits the regression data is their goal.

iii) Logical models: Logical models use a logical expression to divide the instance space into segments and hence construct grouping models. A logical expression is an expression that returns a Boolean value, i.e., a True or False outcome. Once the data is grouped using a logical expression, the data is divided into homogeneous

groupings for the problem we are trying to solve. For example, for a classification problem, all the instances in the group belong to one class.

There are mainly two kinds of logical models: **Tree models** and **Rule models**.

- **Rule Models:** Rule models are a type of logic model that represents decision rules in the form of "if-then" statements. These rules typically involve conditions based on the values of input features and corresponding predictions or decisions. Examples of rule-based models include decision trees, rule induction algorithms and expert systems.
- **Tree Models:** Tree models can be seen as a particular type of rule model where the if parts of the rules are organized in a tree structure. Both Tree models and Rule models use the same approach to supervised learning.
- **Example: Decision Trees**
 - If a customer's income > \$50,000 and credit score > 700, approve the loan; otherwise, reject.
 - Used in expert systems and rule-based classifiers.

iv) Grouping and Grading Models

Grouping and grading are important concepts in machine learning, particularly in the context of classification tasks. The key difference between grouping and grading models is the way they handle the instance space.

Grouping models:

- Grouping also known as clustering is the process of dividing a dataset into clusters or groups of similar data points based on certain criteria or features.
- Clustering algorithms aim to identify natural groupings or patterns in the data without prior knowledge of class labels.
- Common clustering algorithms include K-means clustering, hierarchical clustering, DBSCAN (Density-Based Spatial Clustering of Applications with Noise), and Gaussian mixture models (GMM).

- Clustering is often used for tasks such as customer segmentation, anomaly detection, and data exploration.

Grading models:

- Grading typically refers to the process of assigning labels or scores to data points based on certain criteria or features.
- In the context of classification, grading involves assigning class labels or categories to input data points based on their features.
- Classification algorithms learn to classify data points into predefined classes or categories based on labeled training data.
- Common classification algorithms include logistic regression, decision trees, random forests, support vector machines (SVM), k-nearest neighbors (KNN), and neural networks.
- Classification is used in a wide range of applications including spam detection, sentiment analysis, image recognition and medical diagnosis.

Grouping vs Grading

S.N.	Grouping	Grading
1	Involves dividing data into clusters or categories based on similarities.	Involves assigning a rank, level, or score to data based on performance, quality, or specific criteria.
2	Unsupervised learning is often used, where the algorithm finds patterns without predefined labels.	Often uses supervised learning , where the model is trained on labeled data.
3	Focuses on finding natural structures in data rather than ranking them.	Focuses on ordering or evaluating data points rather than grouping them.

4	Examples: • Clustering: K-Means clustering groups customers based on purchasing behavior. • Categorization: Classifying emails as spam or non-spam.	Examples: • Regression: Predicting house prices based on features (e.g., square footage, location). • Ranking: Search engines ranking webpages based on relevance. • Classification with ordered labels: Assigning letter grades (A, B, C) to student performance.
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1.6 Designing a learning system

Designing a learning system in machine learning (ML) involves several key steps and considerations to ensure that the system effectively learns from data and makes accurate predictions or decisions.

- **Define objectives:**

- ✓ Clearly define the objectives and goals of the learning system. Understand the problem you're trying to solve, the desired outcomes, and how the ML system will be used in practice.

- **Data collection and preprocessing:**

- ✓ Identify and collect relevant data that will be used to train and evaluate the ML model.
- ✓ Clean and preprocess the data to handle missing values, outliers, noise and inconsistencies. This may involve tasks such as data cleaning, feature engineering, feature scaling and dimensionality reduction.

- **Model selection:**

- ✓ Choose an appropriate ML model or ensemble of models based on the problem requirements, data characteristics, and computational resources.
- ✓ Consider factors such as model complexity, interpretability, scalability, and performance metrics.
- **Split data into training and evaluation sets:**
 - ✓ Split the data into training, validation, and test sets to assess the performance of the ML model.
 - ✓ Use the training set to train the model, the validation set to tune hyper parameters and evaluate model performance during training, and the test set to evaluate the final model's generalization performance.
- **Train the model:**
 - ✓ Train the selected ML model using the training data. This involves optimizing the model parameters or weights to minimize a loss function, typically through techniques such as gradient descent or its variants.
 - ✓ Monitor the training process to ensure convergence and prevent overfitting by using techniques such as early stopping and regularization.
- **Evaluate model performance:**
 - ✓ Evaluate the trained model's performance using the validation set or cross-validation techniques.
 - ✓ Use appropriate evaluation metrics based on the problem type (e.g., accuracy, precision, recall, F1-score for classification; mean squared error, R-squared for regression).
 - ✓ Analyze the model's strengths, weaknesses, and areas for improvement.
- **Fine-tune and optimize:**
 - ✓ Fine-tune hyper parameters and model architecture based on the validation results to improve performance.

- ✓ Experiment with different algorithms, feature representations, and preprocessing techniques to find the best-performing configuration.
- **Deploy and monitor:**
 - ✓ Deploy the trained model into production or real-world applications.
 - ✓ Monitor the model's performance over time and retrain it periodically with new data to maintain accuracy and relevance.
 - ✓ Implement mechanisms for error detection, model explain ability, and performance tracking to ensure robustness and reliability.
- **Iterate and Improve:**
 - ✓ Continuously iterate on the learning system based on feedback, new data, and evolving requirements.
 - ✓ Regularly evaluate and update the ML model to adapt to changes in the data distribution or environment.

1.8 Machine learning frameworks

Machine learning frameworks are software tools that simplify the development process of machine learning models by providing libraries and APIs for tasks like data preprocessing and model building. There are various reason for learning ML frameworks, because it simplifies the complex tasks of machine learning and also save the development time.

Tensor Flow:

- Tensor Flow is a free, end-to-end open-source platform developed by the Google Brain team. It offers a wide variety of tools, libraries, and resources for machine learning.
- We can easily build and train machine learning models using high-level APIs such as Keras within Tensor Flow.
- Tensor Flow provides multiple levels of abstraction, allowing you to choose the option that best suits your model.

- It supports deployment of models anywhere, including the cloud, browser, or your own device.
- Available for Python and C APIs, as well as other languages like C++, Java, JavaScript, Go, and Swift.
- Tensor Flow Extended (TFX) provides a full experience, Tensor Flow Lite is for mobile devices and TensorFlow.js allows training and deployment in JavaScript environment.

Theano:

- Theano is an open-source Python library that allows manipulation and evaluation of mathematical expressions, especially those involving multidimensional arrays.
- While it has been less popular in recent years, it was historically used for deep learning research and development.

Scikit-Learn:

- Scikit-learn is a Python library that provides simple and efficient tools for data mining, machine learning, and data visualization.
- It includes various algorithms for classification, regression, clustering, dimensionality reduction and more.
- Scikit-learn is widely used for its ease of use and extensive documentation.

PyTorch:

- PyTorch is an open-source machine learning framework developed by Facebook's AI Research lab.

- It is known for its dynamic computation graph, making it popular among researchers and practitioner.
- PyTorch provides flexibility and ease of debugging, making it a preferred choice for deep learning projects.

Caffe:

- Caffe is a deep learning framework developed by the Berkeley Vision and Learning Center (BVLC).
- It is particularly well-suited for convolutional neural networks (CNNs) and image classification tasks.
- Although it has been around for a while, its usage has decreased in recent years due to the rise of other frameworks.

Stages for building machine learning system

Problem definition and goal setting:

- ✓ Clearly define the problem you want to solve and set specific, measurable goals for the ML system.
- ✓ Understand the business or application requirements, constraints and success criteria.

Data collection and exploration:

- ✓ Identify and collect relevant data sources that will be used to train and evaluate the ML model.
- ✓ Explore and analyze the data to understand its characteristics, distributions, and potential biases.
- ✓ Perform data cleaning, preprocessing, and feature engineering to prepare the data for modeling.

Model selection and development:

- ✓ Choose appropriate ML algorithms and techniques based on the problem type, data characteristics, and objectives.
- ✓ Split the data into training, validation, and test sets for model development and evaluation.
- ✓ Develop and train the ML model using the training data, optimizing model parameters and hyper parameters as needed.
- ✓ Evaluate the model's performance using appropriate evaluation metrics and iterate on the model architecture or parameters as necessary.

Model evaluation and validation:

- ✓ Validate the trained model using the validation set or cross-validation techniques to assess its generalization performance.
- ✓ Perform thorough model evaluation, analyzing its strengths, weaknesses and areas for improvement.
- ✓ Compare the performance of different models and select the best-performing one for deployment.

Deployment and integration:

- ✓ Deploy the trained model into production or real-world applications, integrating it with existing systems or workflows.
- ✓ Implement mechanisms for model versioning, monitoring, and error handling to ensure reliability and scalability.
- ✓ Develop APIs or interfaces for accessing and interacting with the deployed model.

Monitoring and maintenance:

- ✓ Monitor the deployed ML system's performance and behavior in production, tracking key metrics and detecting anomalies or drift.
- ✓ Establish processes for retraining the model periodically with new data to maintain accuracy and relevance.

- ✓ Continuously evaluate and update the ML system to address changing requirements, data distributions, or environmental factors.

Documentation and communication:

- ✓ Document each stage of the ML system development process, including data sources, preprocessing steps, model architecture, and deployment procedures.
- ✓ Communicate findings, insights, and results to stakeholders, domain experts and end-users in a clear and understandable manner.
- ✓ Provide documentation, tutorials, and user guides to facilitate adoption and usage of the ML system.

Machine Learning